HANOI UNI. OF SCI. AND TECH. FACULTY OF MATHEMATICS AND INFORMATICS

WRITING TEST, Summer 2025 **Course: Formal Languages and the Theory of Computation** **Course Id: CS3372, Duration:** 60 **minutes**	**Student name:** ..... **Student Id:** ..... **No.** ..... **Class Id:** .....

Assessors' name and signature	**Proctors' name and signature**	**Total score**

Test Id: 1 **(This test consists of 10 questions)**

Note: ***No document is allowed.***

Questions with one correct answer

Question 1. Let $\Sigma = \{0, 1, \ldots, 9\}$. Which of the following languages is not regular?

- ☐ $0 \cup \{1\}\Sigma^* \cap \Sigma^*\{0, 2\}$
- ☐ $\{0, 1\}^*$
- ☐ $0 \cup \{1, 2, \ldots, 9\}\Sigma^*$
- ☐ $\{0^n 1^n : n \geq 0\}$

Question 2. Which of the following languages is a finite language?

- ☐ $0 \cup \{1, 2, \ldots, 9\}^*$
- ☐ $\{0, 1\}^*$
- ☐ $\{0^n 1^n : 1 \leq n \leq 10\}$
- ☐ $\{0^n 1^n : n \geq 0\}$

Question 3. Let $G = (V, \Sigma, R, S)$ be a context-free grammar, where $V = \{S, E, N, D\}$, $\Sigma = \{0, 1, \ldots, 9, (,), *, +\}$ and R consists of the rules:

$$\begin{array}{l} S \longrightarrow E \\ E \longrightarrow E + E \mid E * E \mid (E) \mid N \\ N \longrightarrow ND \mid D \\ D \longrightarrow 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9 \end{array}$$

Which of the following strings is generated by G?

- ☐ $2 * 3 * 4 - 5$
- ☐ $0 + 01 * 2$
- ☐ $00 + 1 * 2+$
- ☐ $x := y + z$

Question 4. Let M be the deterministic finite automaton $(Q, \Sigma, \delta, q_0, F)$, where $Q = \{q_0, q_1\}$, $\Sigma = \{a, b\}$, $F = \{q_0\}$ and $\delta(q_0, a) = q_0$, $\delta(q_0, b) = q_1$, $\delta(q_1, a) = q_1$, $\delta(q_1, b) = q_0$. Which of following strings is accepted by M?

- ☐ $baaa$
- ☐ $abbab$
- ☐ ϵ
- ☐ $aabba$

Question 5. Consider the pushdown automaton $M = (Q, \Sigma, \Gamma, \delta, q_0, F)$, $Q = \{s, f\}$, $\Sigma = \{a, b\}$, $\Gamma = \{a\}$ and δ contains the five transitions: $((s, a, \epsilon), (s, a))$, $((s, b, \epsilon), (s, a))$, $((s, a, \epsilon), (f, \epsilon))$, $((f, a, a), (f, \epsilon))$, $((f, b, a), (f, \epsilon))$. Which of the following strings is accepted by M?

- ☐ aaa
- ☐ baa
- ☐ aba
- ☐ abb

Question 6. Consider the Turing machine $M = (Q, \Sigma, \delta, q, \{h\})$, $\Sigma = \{a, b, \sqcup, \triangleright\}$. Which of the following pairs is a configuration of M?

- ☐ $(q, \triangleright \underline{\sqcup} baa)$
- ☐ $(q, \triangleright \underline{a} ba \sqcup \triangleright)$
- ☐ $(q, \sqcup \triangleright a \underline{\sqcup} ba)$
- ☐ $(h, \triangleright \underline{a} ba \sqcup)$

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Constructed-response questions

Question 7. Draw the state diagram for a nondeterministic finite automaton that accepts the language: $(ab)^*(ba)^* \cup bb$.

Question 8. Give the regular expression generating the language $L = \{ab, ba, aa\}^*$.

Question 9. Let $G = (V, \Sigma, R, S)$ be a context-free grammar, where $V = \{S, A\}$, $\Sigma = \{a, b\}$ and R consists of the rules $S \to AA$, $A \to a$, $A \to b$, $A \to aA$ and $A \to bA$. Identify $L(G)$.

Question 10. Consider the Turing machine $M = (Q, \Sigma, \delta, s, H)$, where $Q = \{q_0, q_1, h\}$, $\Sigma = \{a, \sqcup, \triangleright\}$, $s = q_0, H = \{h\}$ and $\delta(q_0, a) = (q_1, \sqcup)$, $\delta(q_0, \sqcup) = (h, \sqcup)$, $\delta(q_0, \triangleright) = (q_0, \rightarrow)$, $\delta(q_1, a) = (q_0, a)$, $\delta(q_1, \sqcup) = (q_0, \rightarrow)$, $\delta(q_1, \triangleright) = (q_1, \rightarrow)$. Trace the computation of M starting from $(q_1, \triangleright \underline{\sqcup} aaaa)$.

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Test Id: 2 **(This test consists of 10 questions)**

Note: ***No document is allowed.***

Questions with one correct answer

Question 1. Let $\Sigma = \{0, 1, \ldots, 9\}$ and let $\omega = 1^k 0^n$, $k > n > 0$. Which of the following languages does not contain ω?

- ☐ $\{0^n 1^n : n \geq 0\}$
- ☐ $0 \cup \{1\}\Sigma^* \cap \Sigma^*\{0, 2\}$
- ☐ $\{0, 1\}^*$
- ☐ $0 \cup \{1, 2, \ldots, 9\}\Sigma^*$

Question 2. Which of the following languages is not a finite language?

- ☐ $0 \cup \{1, 2, \ldots, 9\}^2$
- ☐ $\{0, 1\}^3$
- ☐ $\{0^n 1^n : 1 \leq n \leq 10\}$
- ☐ $\{0^n 1^n : n \geq 0\}$

Question 3. Let $G = (V, \Sigma, R, S)$ be a context-free grammar, where $V = \{S, E, N, D\}$, $\Sigma = \{0, 1, \ldots, 9, (,), *, +\}$ and R consists of the rules:

$$\begin{array}{l} S \longrightarrow E \\ E \longrightarrow E + E \mid E * E \mid (E) \mid N \\ N \longrightarrow ND \mid D \\ D \longrightarrow 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9 \end{array}$$

Which of the following strings is generated by G?

- ☐ $(09 * 01 * 3)$
- ☐ $(00 + 1) * 2+$
- ☐ $2 * 3 * (4 - 5)$
- ☐ $x := (y + z)$

Question 4. Let M be the deterministic finite automaton $(Q, \Sigma, \delta, s, F)$, where $Q = \{s, f\}$, $\Sigma = \{0, 1\}$, $F = \{s\}$ and $\delta(s, 1) = s$, $\delta(s, 0) = f$, $\delta(f, 1) = f$, $\delta(f, 0) = s$. Which of following strings is accepted by M?

- ☐ 0111
- ☐ 11001
- ☐ ϵ
- ☐ 1000

Question 5. Consider the pushdown automaton $M = (Q, \Sigma, \Gamma, \delta, q_0, F)$, $Q = \{s, f\}$, $\Sigma = \{a, b\}$, $\Gamma = \{a\}$ and δ contains the five transitions: $((s, a, \epsilon), (s, a))$, $((s, b, \epsilon), (s, a))$, $((s, a, \epsilon), (f, \epsilon))$, $((f, a, a), (f, \epsilon))$, $((f, b, a), (f, \epsilon))$. Which of the following strings is accepted by M?

- ☐ aaa
- ☐ abb
- ☐ aba
- ☐ bab

Question 6. Consider the Turing machine $M = (Q, \Sigma, \delta, q, \{h\})$, $\Sigma = \{a, b, \sqcup, \triangleright\}$. Which of the following pairs is a configuration of M?

- ☐ $(h, aba \sqcup \underline{\sqcup})$
- ☐ $(q, \triangleright \underline{a} ba \sqcup)$
- ☐ $(q, \triangleright \sqcup baa \underline{\sqcup})$
- ☐ $(q, \sqcup \triangleright a \underline{\sqcup} ba)$

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Constructed-response questions

Question 7. Draw the state diagram for a nondeterministic finite automaton that accepts the language: $(bba)^* \cup (abb)^*$.

Question 8. Give the regular expression representing the language $L = \{ba, b, bb, a\}^*$.

Question 9. Let $G = (V, \Sigma, R, S)$ be a context-free grammar, where $V = \{S, B\}$, $\Sigma = \{0, 1\}$ and R consists of the rules $S \to BBB$, $B \to 0$, $B \to 1$, $B \to B1$ and $B \to B0$. Identify $L(G)$.

Question 10. Consider the Turing machine $M = (Q, \Sigma, \delta, s, H)$, where $Q = \{q_0, q_1, h\}$, $\Sigma = \{b, \sqcup, \triangleright\}$, $s = q_0, H = \{h\}$ and $\delta(q_0, b) = (q_1, \sqcup)$, $\delta(q_0, \sqcup) = (h, \sqcup)$, $\delta(q_0, \triangleright) = (q_0, \rightarrow)$, $\delta(q_1, b) = (q_0, b)$, $\delta(q_1, \sqcup) = (q_0, \rightarrow)$, $\delta(q_1, \triangleright) = (q_1, \rightarrow)$. Trace the computation of M starting from $(q_1, \triangleright\underline{\sqcup}bbbb)$.